School of Electrical Engineering and Computer Science
CSI5155 – Machine Learning
Project Fall 2026

Topic: Deep Learning for AI Image Detection

The project may be completed individually or in groups of two or three. Groups are not required to do more work than individuals. Members of a group will all receive the same grade, other than in exceptional circumstances.

Important deadlines:

- Submit your code and final report on Brightspace by **November 24**.
  Note: We cannot accept submissions that are more than 24 hours late (unless you have been given written permission from the instructor).
- In-person lightning talks (short presentations) in class **December 2.** (Pleae upload your slides to Brightspace by the night before.)

100 marks (30% of final grade)
Written report (50 marks)
Implementation (30 marks)
Lightning talk (20 marks)

Project Description

When you see an image on the internet, how can you tell if it is real .. or AI? With today's advanced image generation models, it's becoming harder and harder to tell. In this project, we will explore a deep learning approach to AI image detection.

The goal of the project is to give you an opportunity to practice applying deep learning to a real world problem, and to apply best practices for training, testing, evaluation, and documentation. The project consists of three phases:

Phase 1: Model Training

In this phase, you will train a convolutional neural network (CNN) from scratch, using the provided data. We will be using the "AI-Image Detection Dataset"[1]. It consists of real

[1] https://huggingface.co/datasets/Shanmuk4622/ai-image-detection-dataset

photographs from the COCO and ImageNet-1k datasets, along with images from various AI image generators**.**

For phase 1 and 2, only use the subset of images generated by Stable Diffusion 1.5. You can download the required images directly from the huggingface repository as follows:

```
from datasets import load_dataset, concatenate_datasets
from huggingface_hub import get_token, login

DATASET = "Shanmuk4622/ai-image-detection-dataset"

if get_token() is None:
    login()  # prompts for your token


def load_sd15_task(split):
    """Real photos (label 0) plus Stable Diffusion 1.5 images (label 1) for one split."""
    real = load_dataset(DATASET, "real", split=split)
    sd15 = load_dataset(DATASET, "sd15", split=split)
    return concatenate_datasets([real, sd15]).shuffle(seed=42)


train = load_sd15_task("train")
val = load_sd15_task("validation")
test = load_sd15_task("test")
```

Note that the repository is gated so you must create a huggingface account and use your access token to download the data.

By default, the images are 512x512 RGB images. To make the computation more tractable, please **downsample the images to 128x128.**

You may design your CNN architecture as you see fit, as long as it takes 128x128 pixel images as input, and outputs a binary label (0 = real, 1 = AI-generated).

In the report, include:

- A description of your CNN architecture and any design decisions you made
- A description of your training procedure, including any hyperparameter optimization
- A table reporting recall, precision, F1, and accuracy on the test set
- A short discussion of any key observations

Phase 2: Transfer learning

In this phase, rather than training from scratch, you will instead use a pretrained backbone model, and fine-tune it to your task. Use the same data as in Phase 1 (Stable Diffusion 1.5 and real images), but now you will use the training data as fine-tuning data.

The pretrained model you will use is a ResNet model trained on ImageNet-1k to do image classification: **resnet18(weights=ResNet18_Weights.IMAGENET1K_V1)**

You may do full fine-tuning, or keep some of the weights frozen: this is a design decision. In general, we would expect performance to increase in Phase 2 over Phase 1.

In the report, include:

- A description of your fine-tuning procedure, including any hyperparameter optimization
- A table reporting recall, precision, F1, and accuracy on the test set
- A short discussion of any key observations: what changed in Phase 2 versus Phase 1?

Phase 3: Shared Task Competition

Phase 3 is where you can get creative. Your goal is to build the **best AI detector you can** (subject to constraints below!). The professor and TA will test your final model on a *held-out test set* (i.e., a set of images you don't have access to). The best-performing detectors will be announced on the day of the presentations (Dec. 2).

Note: where your model lands in the final ranking does not impact your grade – it is just a friendly competition! Instead, you will be graded on your ideas, effort, and communication and analysis of results.

Some ideas to consider in this phase:

- So far, the model has only been trained to detect images generated by Stable Diffusion 1.5. To help your model generalize, use some of the other data available in https://huggingface.co/datasets/Shanmuk4622/ai-image-detection-dataset (e.g. generated by FLUX, Kandinsky, and other AI image generators). Be sure to down-sample the images to 128x128 resolution.
- You may also want to construct your own test set to test for generalizability, to simulate the hidden test set.
- You can read some of the published papers on this problem to get ideas for how to improve your detector.
- The held-out test data will be 128x128 real and generated images.

Important Constraints!

- You **may not** use any API calls (e.g. to commercial AI models).
- Your final model must run on a Google Colab T4 GPU in 10 minutes or less for 5000 images.
- All required packaged must be specified in a requirements.txt file

We will provide a verification notebook so you can verify that your model runs as expected before submission. It will run on the free tier of Google Colab. Please include a screenshot that your model passed the verification check in the final report.

In the report, include:

- Description of your approach to improve the model's accuracy and generalizability
- A table reporting recall, precision, F1, and accuracy on the original test set (from phase 1-2) as well as on any new test sets that you constructed to test generalizability
- A short discussion of any key observations: what worked? What didn't? What would you have liked to try if you had more compute budget?

Project Report [50%]

The final report should be approximately 6-8 pages of text (with extra space allowed for figures and references) and structured as follows:

- Brief statement explaining how each group member contributed to the project
- Introduction
- Phase 1
- Phase 2
- Phase 3
- Conclusion
- References (if needed)

Your report will be assessed on its clarity and completeness, as well as the validity of your results and analysis.

Project Implementation [30%]

Projects must be implemented in Python. It is highly recommended that you use PyTorch for your implementation. A template has been provided on Brightspace to ensure your submission is correctly formatted: **please use it.** In particular,

test_submission.py will validate that your submission is correctly formatted for the held-out testing and will run in a reasonable amount of time. After you run test_submission.py it will create a zip file that you can submit.

Submit the zip file containing your code on BrightSpace in the format specified by the template. Your code will be assessed on its functionality, correctness, organization, and adherence to the constraints described above.

If needed, you can run your experiments using the Morning Star compute resources (or other resources such as Google Colab).

Please ensure that you modify requirements.txt to include any additional packages your project needs.

Lightning Talk [20%]

A “lightning talk” is a short presentation, often given at conferences. The goal of a lightning talk is to highlight one or two main takeaways from your work, rather than giving all the details of your project. In this case, since all groups are working on the same project, you do not need to explain the project goals or the dataset. Instead, focus on phase 3 and **any interesting observations, innovations, or failure modes** that you want to share with the class.

Each group will be given a maximum of 3 minutes to present, followed by 2 minutes of questions. You may choose one group member to do the presentation, or multiple group members. Due to the size of the class, time will be strictly enforced!

If you would like to use slides, please upload them to Brightspace by the night before (Dec 1), so they can be compiled into a single deck.

Reminder of our class AI Policy:

Students may use AI tools for basic word processing functions, including grammar and spell checking (e.g. Grammarly, Microsoft Word Editor, Copilot). Students may also use AI tools for clarifying challenging concepts, understanding error messages, generating study questions, or brainstorming. Such uses of AI do not need to be declared.

Students are *not* permitted to use AI to generate entire assignment solutions, write code from scratch, or generate reports from scratch. *The first draft should always be your own.* If AI is then used to substantially modify or improve your first draft, then please declare it (the name of the tool and how it was used) in your assignment/report. You will not be penalized for declaring the use of AI. You must still be able to explain the content of your assignments and project if asked by the instructor or TA.